

AARYN BIRO

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PROFESSIONAL SUMMARY

Senior software engineer and founder with 8+ years of experience building production backend APIs, consumer-facing applications, and cloud platforms across Next.js, Python, SQL, React, Swift, and mobile ecosystems. Strong background in architecture, distributed systems, testing, infrastructure automation, and secure product development, with hands-on ownership of scheduling, billing, deployment, observability, and orchestration systems.

PROFESSIONAL EXPERIENCE

Founder & Principal Engineer — Xcelsior 2025 – Present

London, ON (Remote)

- Architected and built a distributed compute platform for GPU workload orchestration, including public APIs, scheduler logic, worker coordination, billing workflows, Next.js product surfaces, and developer-facing tooling.
- Designed secure backend architecture for admission-gated workers, private mesh networking, container isolation, audit-oriented event handling, and trust/compliance controls across a multi-node platform.
- Built modular backend services and a test-focused engineering workflow spanning routing, scheduling, billing, privacy, persistence, CI, and production-minded deployment practices.

Founder & Principal Engineer — PixelEnhance Labs Jan 2024 – Present

London, ON (Remote)

- Architected and delivered a production AI media platform with FastAPI, PostgreSQL, Redis, Celery, React, and Linux-based deployment infrastructure, owning backend services end to end.
- Built asynchronous processing and orchestration layers for GPU-intensive workloads, including queue-based execution, autoscaling, cloud routing, and provider-aware scheduling across hybrid infrastructure.
- Implemented deployment automation, observability, and environment management across Docker, Nginx, and cloud GPU environments to improve reliability and release velocity.

Software Developer — Snappy Inc. Mar 2023 – Dec 2023

Toronto, ON (Remote)

- Delivered product features across iOS and Android application surfaces, supporting reliable release workflows and ongoing iteration within a distributed engineering team.
- Implemented end-to-end membership, rewards, sign-in, and verification flows with attention to usability, resilience, and release quality.

Software Developer — Neurologn Technologies Inc. Jun 2021 – Dec 2022

Toronto, ON (Remote)

- Built mobile features using Flutter and Dart, applying modern state management, asynchronous programming, and robust API integration patterns.
- Contributed across CI/CD, monitoring, analytics, REST API integration, and frontend/backend functionality using Node.js, MongoDB, Express, and React.

Full Stack Software Engineer — realxdata GmbH May 2018 – Sep 2019

Berlin, Germany (On-site)

- Developed production SaaS features using React, TypeScript, Python, and PostgreSQL in a commercial data platform.
- Built backend and frontend functionality for data-driven workflows, reporting, and customer-facing product capabilities, and integrated AWS services and third-party APIs.
- Improved engineering quality through automated testing and CI/CD workflows, accelerating delivery and feedback cycles.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Swift, Dart, Kotlin

Backend: FastAPI, Node.js, PostgreSQL, Redis, MongoDB, REST APIs, background jobs, service-oriented architecture

Frontend & Mobile: Next.js, React, TypeScript, SwiftUI, Flutter, Tailwind CSS

Infrastructure: Docker, Linux, Nginx, AWS, CI/CD, observability, cloud deployment

Systems & AI: PyTorch, TensorRT, FFmpeg, model benchmarking, inference pipelines, GPU orchestration

EDUCATION

Bachelor of Engineering Science, Software Engineering — Western University 2014 – 2020

Honors: Dean's Honor List • MacKay-Lassonde Award in Computer Engineering • International Learning Award

CERTIFICATIONS & SPEAKING

- Cross-platform Mobile Development Summit Speaker — Geekle.us
- Leadership Effectiveness Training (L.E.T.) Certificate — Gordon Training International
- Deloitte Entrepreneurship Bootcamp